Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/01/2015
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Project Title:	Hypobaric Plant Biology - Molecular Responses of Arabidopsis to the Low Atmospheric Pressures of Spaceflight Vehicles and Planetary Habitats		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (University of Florida)		
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Performance Goal Text:			
	Atmospheric pressure and composition are among the engineering variables considered in the design and construction of spaceflight vehicles and extraterrestrial habitats. Simply put, the costs of maintaining a pressure vessel at one atmosphere have been traded away throughout the history of spaceflight vehicle design and are traded away in future designs, yet the biological impacts of lower atmospheric pressure, especially on plants, is both profound and poorly understood. The objective of this proposal is to develop a refined understanding of the metabolic processes involved in plant responses and physiological adaptations to low pressure environments relevant to space exploration. The long-term goal of this line of research is a fundamental understanding of low pressure plant biology within exploration vehicles and structures, with a practical goal of producing plants that are specifically designed to thrive in low atmospheric pressures. The essential drivers of this project are the inescapable engineering limitations to producing orbital, lunar, or		

Task Description:	Martian plant growth facilities that contain Earth-normal atmospheric pressures, and the knowledge that plants do mount complex, and sometimes unexpected costly metabolic responses to hypobaria. These responses can also be used as a model that will allow fundamental understanding of how plants adapt to changing terrestrial environments. In the initial phases of the project, we will use the tools of functional genomics in Arabidopsis plants to examine molecular genetic responses that occur during initial and chronic exposure to atmospheres ranging from Earth normal to near Martian pressures and containing various gas mixtures, especially those atmospheres that are in future vehicle designs. We will build on our previous molecular work using Arabidopsis (Paul et al., 2004). We will conduct extensive tests of a range of appropriate pressures and compositions using wild type genotypes. Then we will use select mutants and engineered changes in metabolic pathways implicated by the initial gene expression array data to test hypotheses regarding the fundamental mechanisms associated with the hypobaria response. Finally, we propose a novel use of International Space Station (ISS) airlock facilities to investigate low pressure processes directly in a spaceflight microgravity environment, to take a first look at the emergent effects of altering atmospheric pressure within an operational vehicle in space. The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial and spaceflight environments. The current proposal for ground-based research leading to spaceflight addresses specifically a focus area defined by the Decadal Study: Plant and Microbial Growth Under Altered Atmospheric Pressures. The proposed research will provide fundamental insights into the biological impact of novel atmospheric environments, a focus area that is specifically identified in the Decadal Study (Decadal-Survey-Commitee 2011).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Experiments in extreme and novel environments, like our spaceflight program and these in low pressure habitats, tell us that that life utilizes special, potentially unique signals to adapt to living with unique challenges. This insight has tremendous implications for the expansion of human existence, beyond, to other worlds, but also richly informs us about the potential for plants to adapt to environmental changes here on Earth. Any insights we reveal have applications to understanding how plants may better cope with some of the stresses we see in our daily environment like drought, global warming, flooding, high CO2, etc. No plant on Earth has ever evolved to respond to ultra-low atmospheric pressures, yet with a little help, plants do just fine in this environment. We are learning what physiological tools plants use to cope with novel stresses, and that teaches us two things. First, it helps us understand what kind of help plants need to be productive on a long spaceflight mission or in a greenhouse on Mars; not an immediate need, but information we need before we do either of those things. Second, how a plant responds to a novel stress gives us a fundamental understanding of how plants cope with an adverse environment. This knowledge can help us predict how plants of the Earth will manage changes in climates that we have not yet seen on a global scale, or how plants might respond to new pollutants. This research is also about understanding how to optimize protected habitats and closed environments, issues that are important to us if we want to grow plants on a space station or long journey, but also if we want to grow plants in a greenhouse on another planetary surface. The lessons we have learned and the tools we have developed growing plants in low pressure chambers, as well as a Martian-analog greenhouse in the arctic, have direct application to protected agriculture in Florida, and anywhere on Earth.
Task Progress:	Plants were grown in the Controlled Environment Systems Research Facility (CSERF) at the University of Guelph. Plants were subjected to a variety of conditons and durations. Plants were grown in two ways: on nutrient agar plates and in rockwool. Plants were harvested into RNAlater for subsequent analysis. I. Plants in Rockwool. The primary goal of this experiment was to address the question of metabolic acclimation to reduced pressures. We compared that with a control of ambient pressure and normal atmospheric composition, and also ambient pressure but with an oxygen atmosphere reduced to the partial pressure equivalent to 10kPa (2% O2). II. Plants on Plates. The plates ran in the chambers at the same time as the plants in Rockwool; however, they were prepared and harvested differently. Both time course and developmental age response experiment were completed during the testing. III. Transcription experiment. Work was completed to evaluate the results of the initial experiments. This includes use of samples described in the experiments above, as well as from past time course and variable pressure experiments. These experiments were analyzed using microarrays to evaluate tissue-specific differential gene expression levels in response to variable hypobaric, and mixed atmosphere treatments.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Bamsey MT, Paul A-L, Graham T, Ferl RJ. " Flexible imaging payload for real-time fluorescent biological imaging in parabolic, suborbital and space analog environments." Life Sci Space Res. 2014 Oct;3:32-44. http://dx.doi.org/10.1016/j.lssr.2014.09.002 , Oct-2014